EART26201 | Principles of Geochemistry | Practical 5

Timescales of magmatic processes FEEDBACK

Task

This practical contains two questions about using radiogenic isotopes to date meteorite formation and track crustal contamination recorded by lavas erupted from an Andean volcano. Each question should take about 60 minutes.

I strongly encourage you to discuss your work with each other as you go.

Demonstrators will be available throughout the practical class, and I will provide a short round-up at the end of the session. Feedback will be posted on Blackboard in due course.

Please share any comments you have as we go along, and I will do my best to adjust.

I suggest typing your answers into this document and/or its associated Excel file.

Goals

At the end of this practical you should be able to:

- Describe the Arrhenian nature of the relationship between temperature and diffusivity
- Explain why different elements in the same crystal will record of different events
- Describe what controls the rate at which crystals achieve thermodynamic equilibrium
- Explain how compositional zoning in crystals records magma mixing timescales
- Apply simple diffusion models to zoned crystals to estimate mixing timescales
- Reconstruct pre-eruptive histories of volcanoes from simple diffusion models

Other themes relevant to this week's content are reviewed in a Blackboard quiz.

You will revisit many of these themes in EART22200 (Igneous Minerals and Processes) and EART24200 (Metamorphic Minerals and Processes) later in the year.

Before you start

Open the file *EART26201_Practical5.xlsx* in Excel.

Question 1: Diffusion coefficients and Arrhenian relationships (60 minutes)

- The diffusivity of an element within a medium (i.e. the speed at which an element diffusion within a given material) is described by the diffusion coefficient (D). The larger the diffusion coefficient, the faster the diffusion. Diffusion coefficients are commonly expressed in units of m^2/s because they relate to the magnitude of the diffusive flux of an element through a plane (Lecture 2 | Part 4).

- The diffusion coefficient of element i (D_i) is related to temperature (T) by the following Arrhenian relationship:

$$D_i = D_0 e^{\frac{-E_a}{RT}},$$

where D_0 is a constant (the 'pre-exponential factor'), E_a is the activation energy and R is the gas constant (Lecture 5 | Part 4).

- The top of the sheet titled *Question 1* contains published data from Chakraborty (1997) who performed high-T experiments to determine the diffusivity of Mg and Fe in olivine (D_{Mg-Fe}). Remember that olivine has the formula $(Mg,Fe)_2SiO_4$ and that Mg and Fe form a solid solution between the Mg-rich endmember forsterite and the Fe-rich endmember fayalite (Lecture 5 | Part 5). This means that the diffusion of Mg and Fe are closely interlinked; one cannot diffuse without exchanging by the other. This kind of diffusion is referred to as interdiffusion; Mg and Fe interdiffuse with each other.

a) Plot the data provided in the sheet titled *Question 1* to illustrate how D depends on T (i.e. plots T as the independent variable on the x-axis and D as the dependent variable on the y-axis). Remember that you should first convert the experimental temperatures from °C into K ($T\ (K) = T\ (°C) + 273.15$). Is it more sensible to use a linear or a logarithmic scale for the y-axis? (Note that axis properties can be changed by clicking on the plot axis and then changing the *Axis Options* visible under *Format Axis*)

Logarithmic; see Excel spreadsheet

b) Right-click on the data and add a trendline. What kind of equation is required to fit the data? (hint: look at the form of the equation given above for diffusion coefficients)

Exponential

c) What is the value of D_{Mg-Fe} at 1350 K?

10^{-17} m^2/s

- Different elements diffuse at different rates. This means that some elements will approach thermodynamic equilibrium by diffusion more quickly than others when the

conditions (i.e. P, T, X) change. Thus, even within a single crystal, different elements will be affected by changes in conditions that took place over different timescales.

d) Demouchy and Maxwell (2006) performed high-T experiments to determine the diffusivity of H in olivine. H is present as a trace element in olivine (see Lecture 3), so we can express its diffusivity as a simple diffusion coefficient without having to consider interdiffusion with other elements like we do for Mg and Fe. Using a D_0 value of $10^{-1.4}$ m^2/s, an E_a value 258 kJ (Excel will show these in a different notation), and the equation given above (remember the minus sign), determine values of D_{H} for an appropriate range of T. R is 8.314 × 10^{-3} kJ/mol/K.

950–1300 °C is appropriate; see Excel spreadsheet

e) Plot your calculated values of D_{H} alongside the values of $D_{\mathrm{Mg-Fe}}$ from Chakraborty (1997). How does the rate of H diffusion in olivine compare with the rate of Mg–Fe interdiffusion? Will H or Mg–Fe reach thermodynamic equilibrium quicker? Given the size and abundance of H, Mg and Fe ions in olivine, explain your answers.

H diffusion is about five orders of magnitude faster than Mg–Fe interdiffusion, so H will attain thermodynamic equilibrium much sooner. H diffuses faster because it is a very small ion (a proton) that can pass through gaps in the olivine crystal lattice quicker that larger ions cannot fit through. As a trace component it also does not have to exchange with lattice-forming species like Mg and Fe; charge balance is achieved by the presence of vacancies, whereby Mg and Fe ions are sometimes missing from the lattice.

f) Ito and Ganguly (2006) performed high-T experiments to determine the diffusivity of Cr in olivine. Like H, Cr is a trace element in olivine, so we can express its diffusivity as a simple diffusion coefficient. Using a D_0 value of $10^{-5.31}$ m^2/s, an E_a value 352 kJ, and the equation given in b), determine values of D_{Cr} for an appropriate range of T.

950–1300 °C is appropriate; see Excel spreadsheet

g) Plot your calculated values of D_{Cr} alongside your calculated values of D_{H} and the values of $D_{\mathrm{Mg-Fe}}$ from Chakraborty (1997). How does the rate of Cr diffusion in olivine compare with the rate of Mg–Fe interdiffusion? Will Cr or Mg–Fe attain equilibrium sooner?

Cr diffusion is about two orders of magnitude slower than Mg–Fe interdiffusion, meaning that Mg–Fe will attain thermodynamic equilibrium before Cr

h) With reference to the above, explain which elements in olivine crystals (i.e. H, Mg–Fe or Cr) are likely to record the following processes taking place beneath active volcanoes:

i. Magma mixing in the shallow crust a few months before eruption — Mg–Fe

ii. H_2O degassing from magmas a few hours before eruption — H

iii. Magma crystallisation in the deep crust tens of years before eruption. — Cr

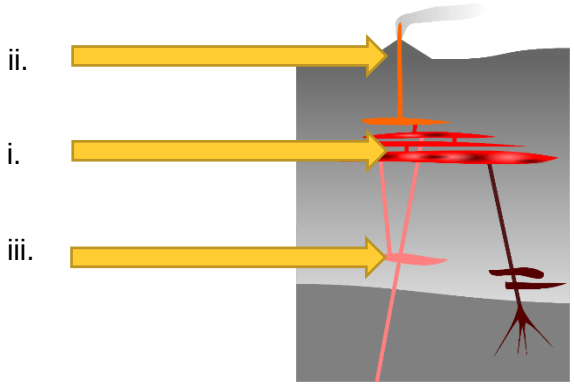


Figure 1: Cartoon of a magma plumbing system

Question 2: Estimating timescales of magma mixing from olivine (60 minutes)

(Thanks to Fidel Costa in Singapore and Maren Kahl in Heidelberg for providing materials)

- Mg–Fe interdiffusion in olivine is commonly used to estimate the timescales over which different magmas mix with each other before volcanic eruptions. Magma mixing in magma chambers brings magmas with different Mg and Fe contents into contact with each other. As a result, the Mg and Fe contents of olivine crystals carried by these magmas progressively homogenise by diffusion, eventually reaching equilibrium.
- If diffusion is interrupted before thermodynamic equilibrium has been achieved (and the diffusion coefficient is known), we can determine the time between when diffusion started and when it was interrupted. In this case, we can determine the time between when the magma mixed and when it cooled during eruption.
- The sheet titled *Question 2 model* contains a simple diffusion model that solves Fick's second law for Mg–Fe interdiffusion in olivine crystals. The calculations relate the composition of zoned olivine crystals expressed in terms of forsterite content (X_{Fo}) along profiles from crystal rims (at 0 µm) to crystal cores (at 250 µm). Olivine crystals may have either lower rim X_{Fo} (so-called normal zoning that reflects cooling before eruption) or higher rim X_{Fo} (so-called reverse zoning that reflects heating before eruption).
- The plot in the sheet titled *Question 2 model* shows how the initially stepped profile of X_{Fo} (the black line) changes in response to diffusion over various lengths of time (shown as different coloured lines). These lengths of time are shown in seconds in the legend. This relaxation of the initial stepped profile is analogous to this visual illustration of diffusion from the lecture (Lecture 5 | Part 4):

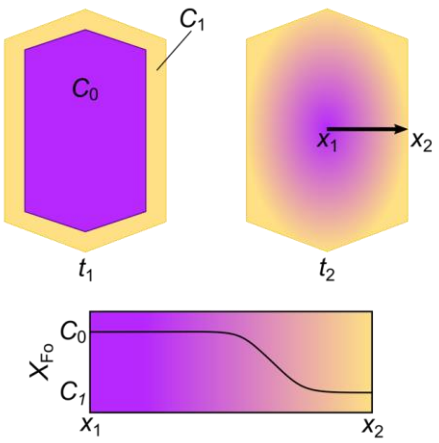


Figure 2: Diffusion in a zoned olivine crystal

a) The shape of the calculated profiles can be changed by editing the green cells *E6* and *E7* that are respectively labelled *Rim* X_{Fo} and *Core* X_{Fo} for the rim and core ends of the profile. Check that you know how to change the values in these cells.

b) Composition profiles from 12 olivines are provided in in the sheet titled *Question 2 data*. Determine the diffusion timescales associated with each of these 12 olivines. Do this by copying one composition profile at a time from *Question 2 data* and pasting the only the values into the pale green cells in column *A* marked *Measured distribution of* $X_{Fo.}$ You can copy values by right clicking on cell *A3* once you've copied the data and then selecting *values* (marked with a little 123) under *paste options*. You can then adjust the values of *Rim* X_{Fo} and *Core* X_{Fo} to overlap the model profiles on top of the measured profile before roughly estimating the timescale by comparing it with the different curves plotted. Note the nature of the composition profile (normal or reverse zoning; see c)), and your timescale estimate for each olivine.

See Excel spreadsheet

c) How many olivines are normally zoned? How many are reversely zoned?

Eight olivines are normally zoned, and four are reversely zoned.

d) How many different olivine populations are there, and what rough timescales are they associated with?

Plotting the internal X_{Fo} of the olivines against the timescales estimated reveals three groups associated with timescales of ~19, ~40 and ~206 hours; see Excel spreadsheet

e) Using your answers to c) and d), reconstruct the history of the magma immediately before eruption. How many different olivine populations were present. In what order did these different populations mix? You can also comment on the relative temperatures of the magma reservoirs from which different olivine populations were supplied using the olivine phase diagram below. Note that the *absolute* temperatures will be incorrect because this phase diagram is for a pure olivine system and you are looking at a basalt.

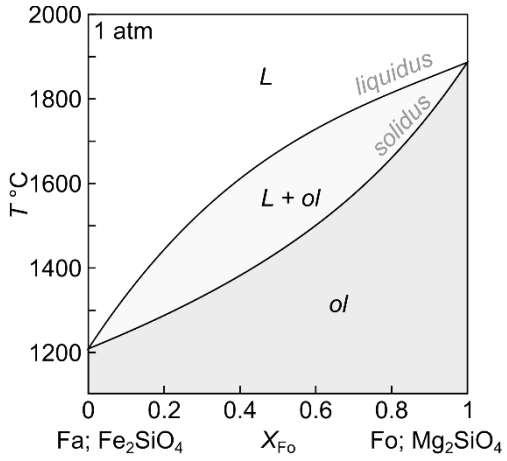


Figure 3: Olivine phase diagram

The first event recorded is the mixing of normally zoned olivine crystals with core X_{Fo} contents of 0.9 into the magma in equilibrium with X_{Fo} contents of 0.7 about 206 hours before eruption. These crystals came from the hottest environment.

We then record the mixing of reversely zoned olivine crystals with core X_{Fo} contents of 0.55 into the magma in equilibrium with X_{Fo} contents of 0.7 about 40 hours before eruption. These crystals came from a cooler environment than the environment in which the crystal rims formed.

Finally, we record the mixing of normally zoned olivine crystals with core X_{Fo} contents of 0.8 into the magma in equilibrium with X_{Fo} contents of 0.7 about 19 hours before eruption. These crystals came from a slightly hotter environment than the environment in which the crystal rims formed.

- Although this exercise simplifies both the complexity of natural magmatic systems and the diffusion modelling process, the principles are directly analogous to how volcanologists use olivine crystals to reconstruct the past history of volcanoes. This figure from Kahl *et al.* (2017) illustrates how olivine crystals recorded magma movements beneath Mt Etna, Italy, before a major eruption on 1669:

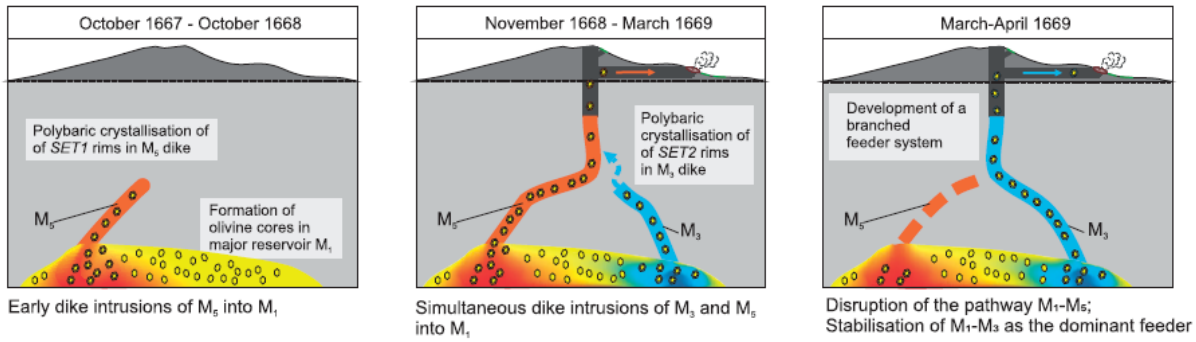


Figure 4: Magma interactions and magma ascent pathways before the 1669 eruption of Mount Etna, Italy

f) Adjust the values given in pale yellow boxes to see how changing the diffusion time and temperature affect the profiles calculated. Note that if you change the values too much you may break the model!

Note how changing the diffusion time changes the length of the profiles, with longer time increments leading to longer, smoother profiles. Long timescales of months to years like those described in Figure 4 are recorded by profiles over 100s um. Eventually the profiles will become flat, but the simple excel model we're using here.

The effect of changing temperature is dramatic. As you linearly increase temperature, calculated diffusion time drop dramatically because diffusion is an Arrhenian process.